

AI-powered Denoising: Unsupervised Learning

The denoising problem

Estimate this



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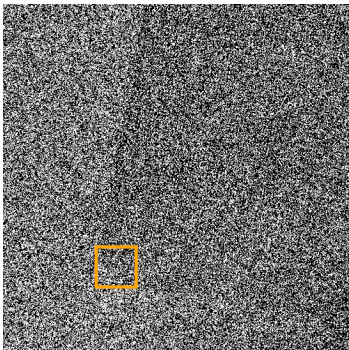


From this



Why?

Why?



Supervised denoising

Supervised denoising

Minimize mean squared error (MSE) on **clean** - **noisy** pairs

$$\left\| \begin{array}{c} \text{MSE} \\ \text{denoised} - \text{clean} \end{array} \right\|_2^2$$

The diagram illustrates the Mean Squared Error (MSE) calculation for supervised denoising. It features two grayscale photographs of a person in a dark coat operating a camera on a tripod. The left photograph is labeled 'denoised' and is covered with a heavy salt-and-pepper noise pattern. The right photograph is labeled 'clean' and is clear. A horizontal minus sign is positioned between the two images. Above the images, the text 'MSE' is centered, and to the right of the images, the notation $\left\| \right\|_2^2$ indicates the squared L2 norm of the difference between the two images.

Supervised denoising

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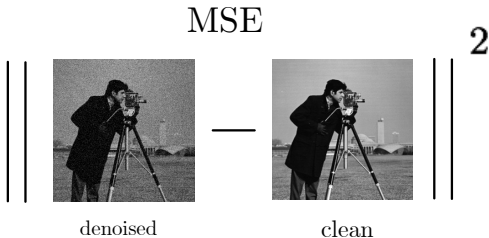
The diagram illustrates the Mean Squared Error (MSE) calculation for supervised denoising. It features two grayscale photographs of a man in a dark coat operating a camera on a tripod. The left photograph is labeled 'denoised' and shows significant salt-and-pepper noise. The right photograph is labeled 'clean' and is clear. A horizontal minus sign is positioned between the two images. Above the images, the text 'MSE' is centered. To the right of the images, a vertical double line is followed by a superscript '2', indicating the squared L2 norm.

Good news:

Bad news:

Supervised denoising

Minimize mean squared error (MSE) on **clean** - **noisy** pairs

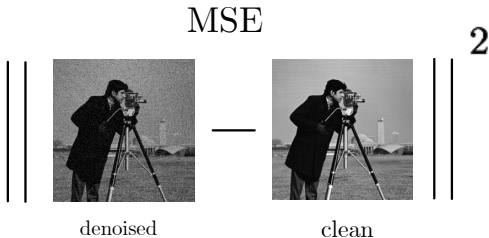
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Good news: Works well!

Bad news:

Supervised denoising

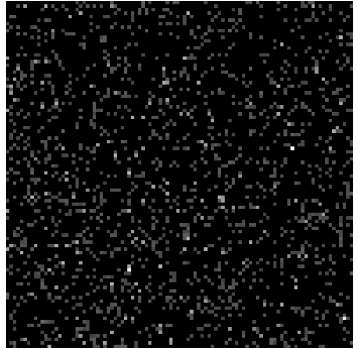
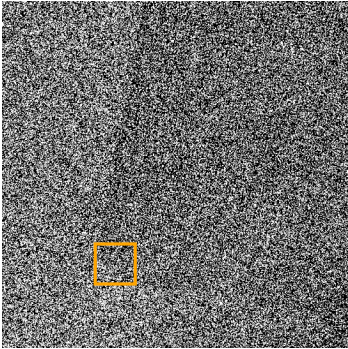
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Good news: Works well!

Bad news: Requires clean images...

... which are often not available in practice



Plan

The Noise2Noise framework

Unsupervised methods

Real data

The Noise2Noise framework

Unsupervised methods

Real data

Unsupervised learning

Main insight: Learn to approximate **noisy target**, instead of clean image



Denoised

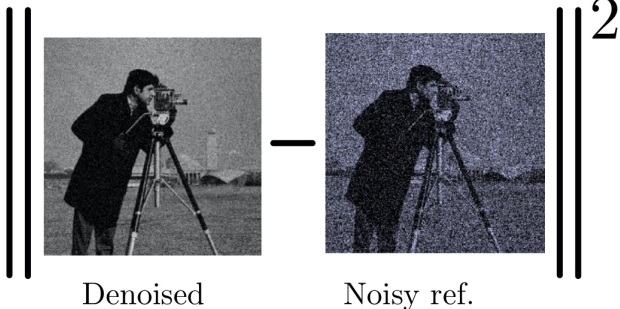


Noisy ref.

2

Unsupervised learning

Main insight: Learn to approximate **noisy target**, instead of clean image



Noise2Noise: Learning image restoration without clean data.
Lehtinen et al. 2018

Additive noise

Clean signal: x Data: $y = x + z$ Denoised estimate: $\mathcal{D}(y)$

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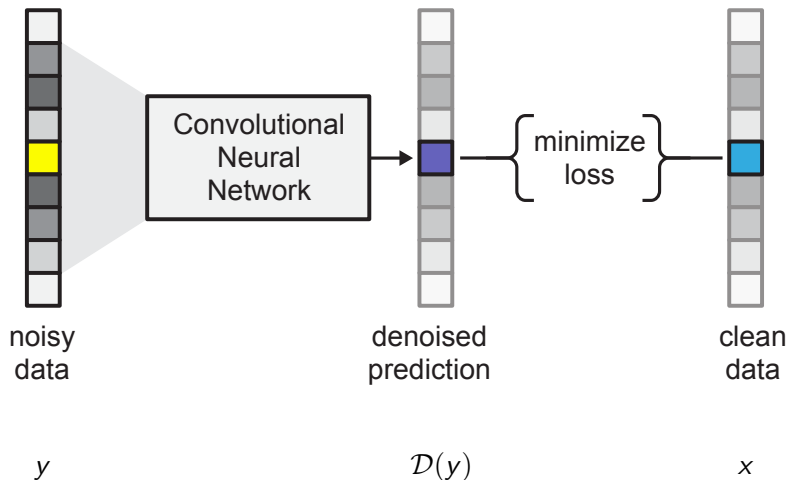
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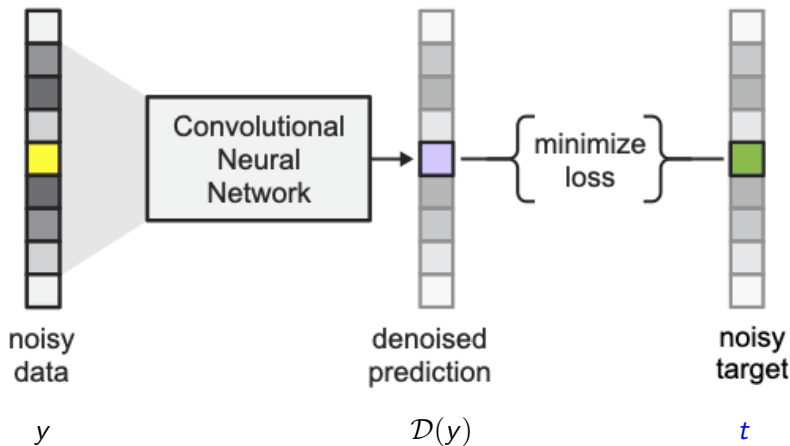
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Supervised learning



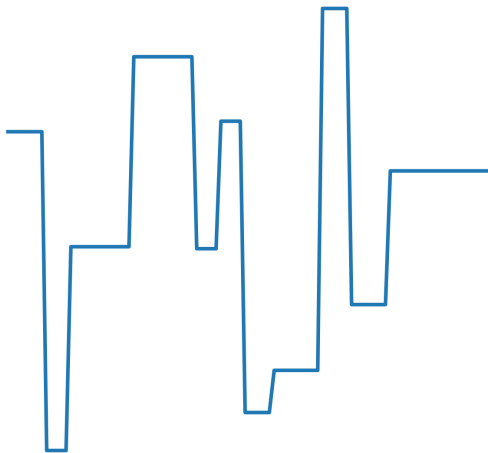
Noise2Noise



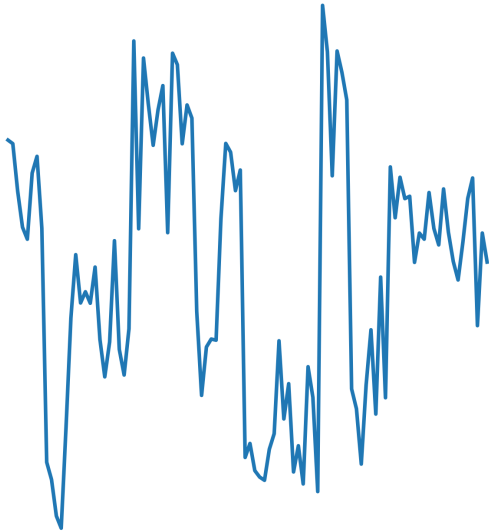
Example datasets

1. 1D piecewise-constant functions
2. Natural photographic images
3. Simulated electron-microscopy data

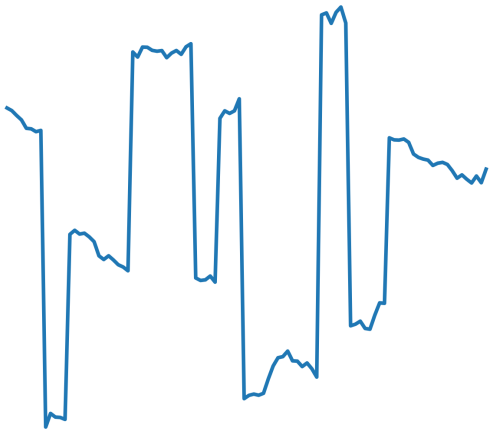
1D piecewise-constant functions



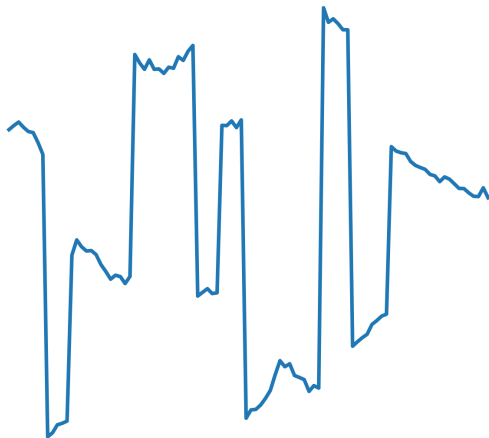
Additive i.i.d. Gaussian noise (PSNR 19.96 dB)



Supervised (PSNR 26.44 dB)



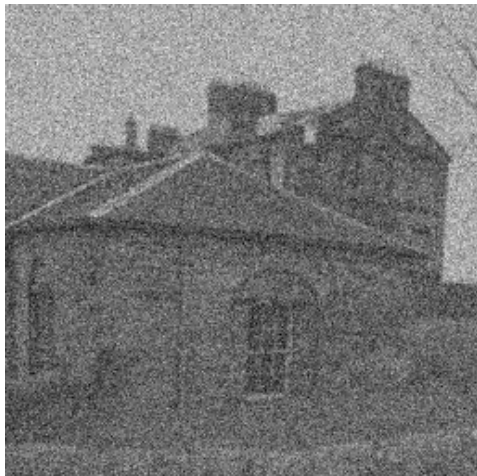
Unsupervised (PSNR 26.30 dB)



Natural images



Additive i.i.d. Gaussian noise with variable variance
(PSNR 19.99 dB)



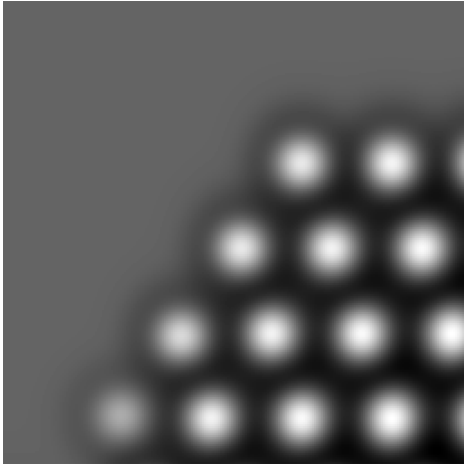
Supervised (PSNR 30.78 dB)



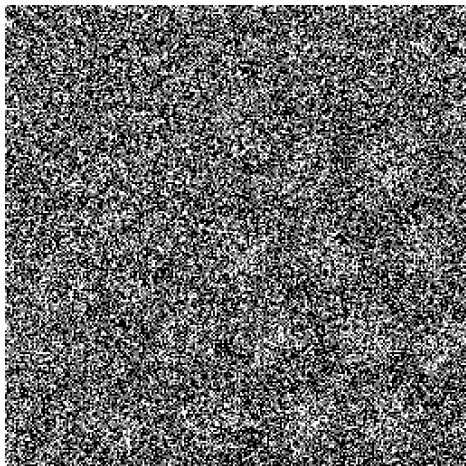
Unsupervised (PSNR 30.03 dB)



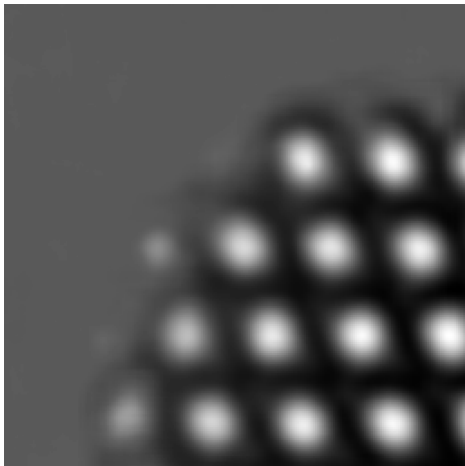
Electron microscopy



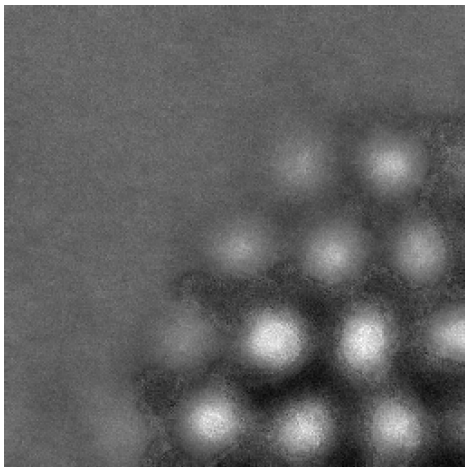
Poisson shot noise (PSNR 6.64 dB)



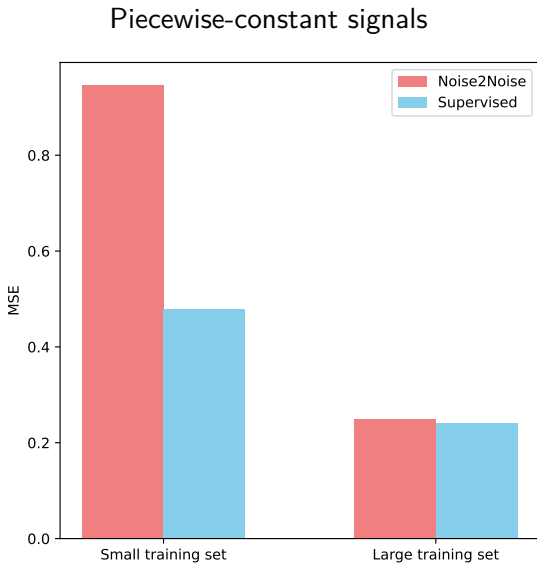
Supervised (PSNR 34.78 dB)



Unsupervised (PSNR 30.82 dB)



Effect of training-set size



Unsupervised evaluation

Clean signal: x Data: $y = x + z$ Denoised estimate: $\mathcal{D}(y)$

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Noisy target: $t = x + w$

If w and z are **independent**

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Can be corrected to *estimate the supervised MSE* and perform **unsupervised evaluation**

Unsupervised MSE

$$\left\| \begin{array}{c} \text{Denoised} \\ \text{Noisy ref. \#1} \end{array} \right\|^2 - \frac{1}{2} \left\| \begin{array}{c} \text{Noisy ref. \#2} \\ \text{Noisy ref. \#3} \end{array} \right\|^2$$

The diagram illustrates the Unsupervised MSE loss function. It shows two image pairs being compared. The first pair consists of a 'Denoised' image and a 'Noisy ref. #1' image. The second pair consists of a 'Noisy ref. #2' image and a 'Noisy ref. #3' image. The loss is calculated as the squared L2 norm of the difference between the first pair, minus half the squared L2 norm of the difference between the second pair.

Unsupervised MSE

$$\left\| \begin{array}{c} \text{Denoised} \\ \text{Noisy ref. \#1} \end{array} \right\|^2 - \frac{1}{2} \left\| \begin{array}{c} \text{Noisy ref. \#2} \\ \text{Noisy ref. \#3} \end{array} \right\|^2$$

The diagram illustrates the Unsupervised Mean Squared Error (uMSE) calculation. It shows two pairs of images. The first pair consists of a 'Denoised' image and a 'Noisy ref. #1' image. The second pair consists of a 'Noisy ref. #2' image and a 'Noisy ref. #3' image. The images are arranged in a row, with the first pair on the left and the second pair on the right. The 'Denoised' image is a clear black and white photo of a person taking a picture. The 'Noisy ref. #1' image is the same photo with added noise. The 'Noisy ref. #2' image is the same photo with added noise and a green tint. The 'Noisy ref. #3' image is the same photo with added noise and a brown tint. The images are separated by minus signs, and the entire expression is enclosed in large double vertical bars with a superscript 2, indicating a squared norm.

$$\text{uMSE} := \frac{1}{n} \sum_{i=1}^n (t_i - \mathcal{D}(y)_i)^2 - \frac{(b_i - c_i)^2}{2}$$

Unsupervised MSE

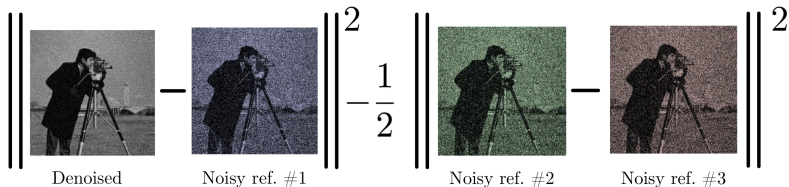
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The diagram illustrates the Unsupervised MSE calculation. It shows a 'Denoised' image (a person with a telescope) and three 'Noisy ref.' images (the same person with different noise patterns). The formula calculates the squared L2 norm of the difference between the denoised image and the first noisy reference, minus half the squared L2 norm of the difference between the second and third noisy references.

$$\text{uMSE} := \frac{1}{n} \sum_{i=1}^n (t_i - \mathcal{D}(y)_i)^2 - \frac{(b_i - c_i)^2}{2}$$

If noisy target t and references b, c correspond to *same clean image* with *independent* noise, uMSE is **unbiased** and **consistent**

Unsupervised MSE



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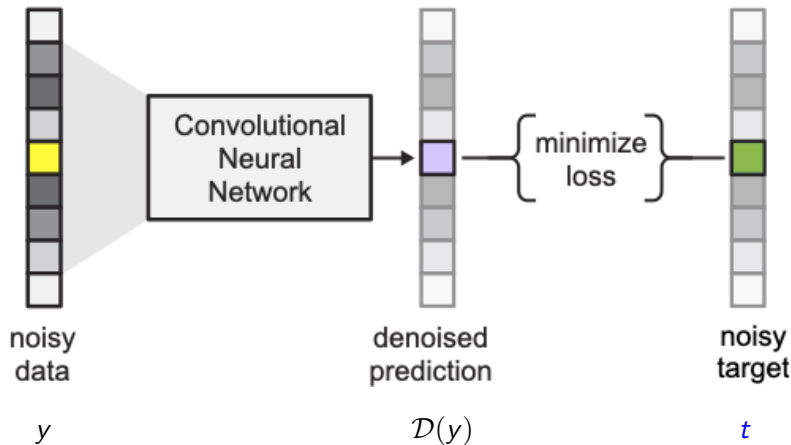
Evaluating unsupervised denoising requires unsupervised metrics.
Marcos-Morales et al. 2023

The Noise2Noise framework

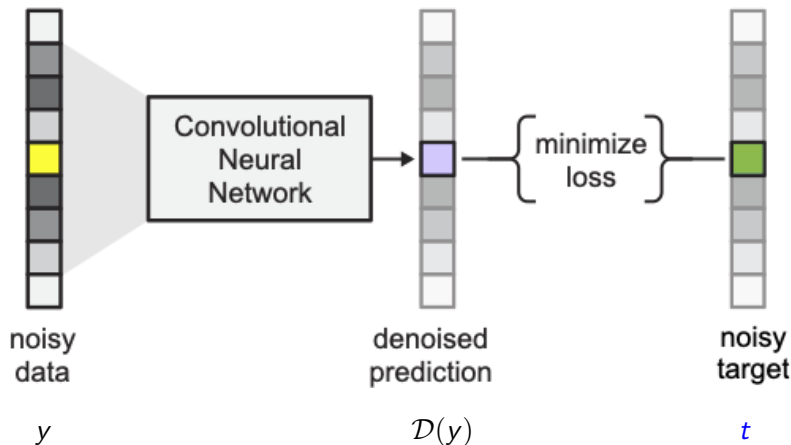
Unsupervised methods

Real data

Noise2Noise framework



Noise2Noise framework



Key challenge: How to obtain the noisy target?

Key challenge

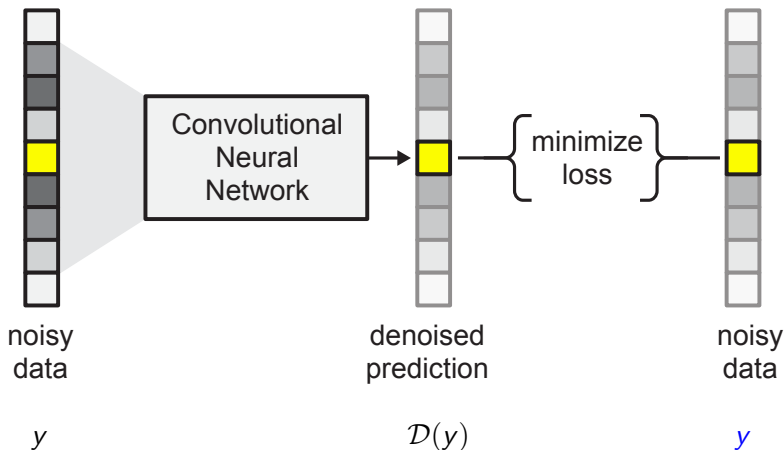
Ideally, target should correspond to the **same signal** as the noisy data, corrupted by **independent noise**

Key challenge

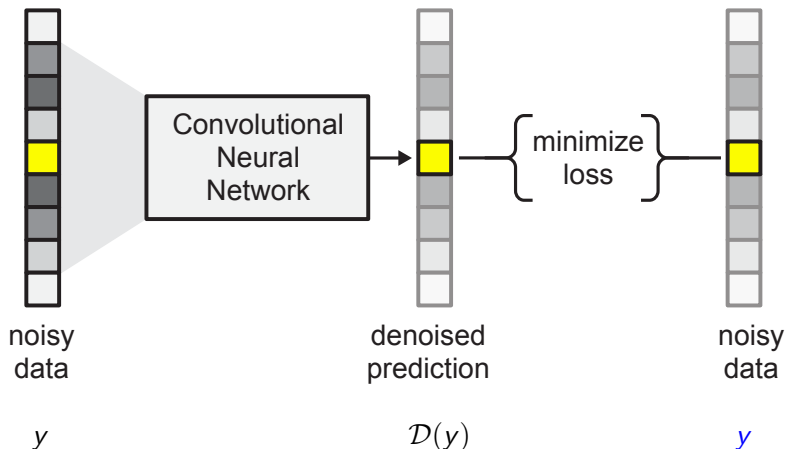
Ideally, target should correspond to the **same signal** as the noisy data, corrupted by **independent noise**

Different strategies to estimate the target result in different unsupervised methods

Strategy 1: Use input noisy data

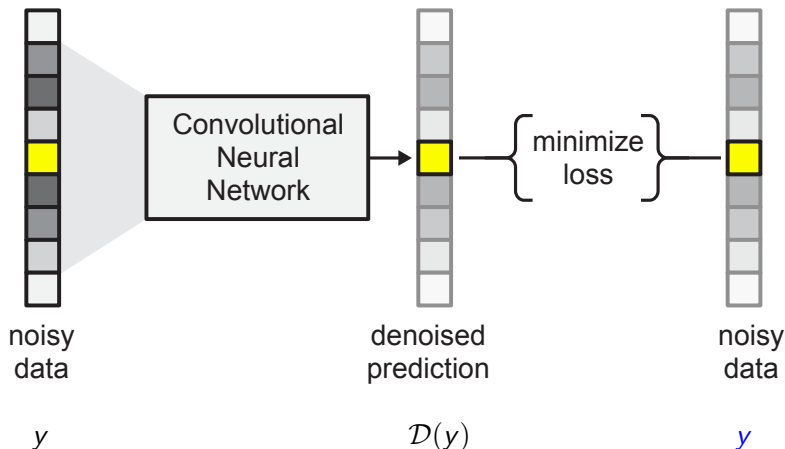


Strategy 1: Use input noisy data



Problem: Non-denoising identity function is optimal

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Problem: Non-denoising identity function is optimal

Can work if model is **constrained** to avoid fitting noise
(*Deep image prior*. Ulyanov et al. 2018)

Strategy 2: Temporal sampling

Use two frames acquired successively as noisy input and target

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Assumptions:

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- ▶ Noise is frame-wise independent

Strategy 2: Temporal sampling

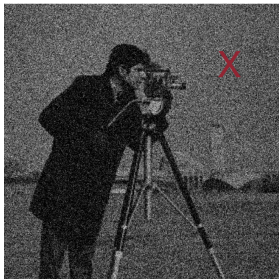
Use two frames acquired successively as noisy input and target

Assumptions:

- ▶ Noise is frame-wise independent
- ▶ Signal changes slowly over time

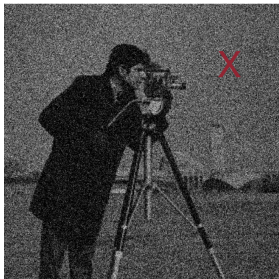
Strategy 3: Spatial sampling

Obtain noisy data and target via **spatial subsampling**



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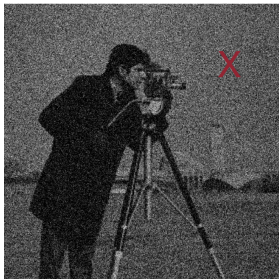
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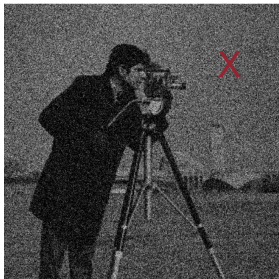


Assumptions:

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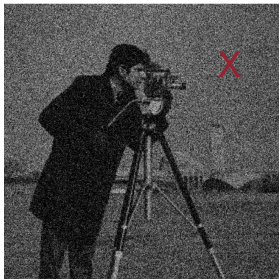


Assumptions:

- ▶ Noise is pixel-wise independent
- ▶ Signal is spatially smooth

Strategy 3: Spatial sampling

Obtain noisy data and target via **spatial subsampling**

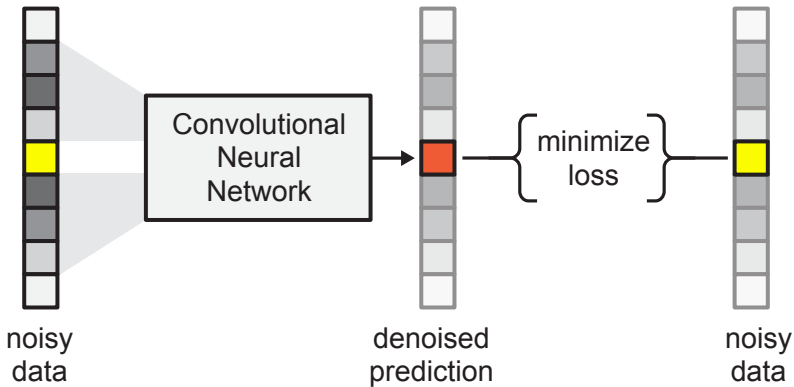


Assumptions:

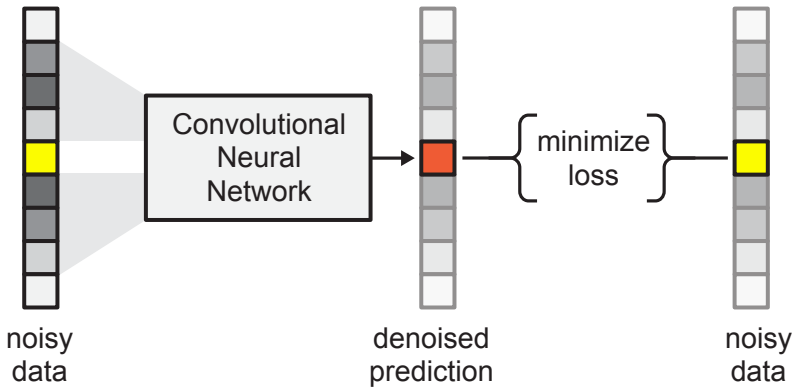
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Neighbor2Neighbor: Self-supervised denoising from single noisy images. Huang et al. 2021

Strategy 4: Blindspot denoising

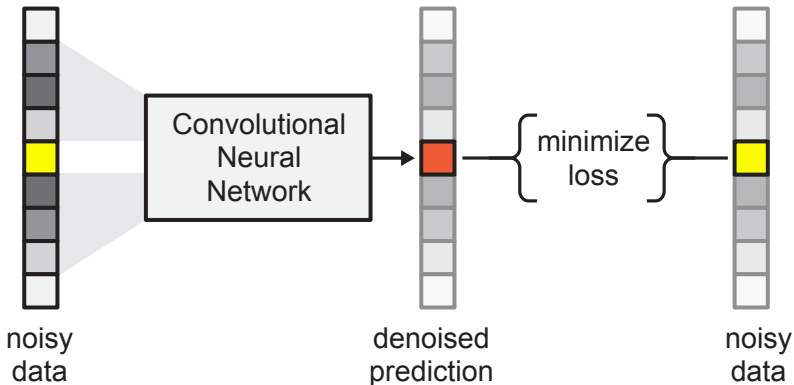


Strategy 4: Blindspot denoising



Assumptions:

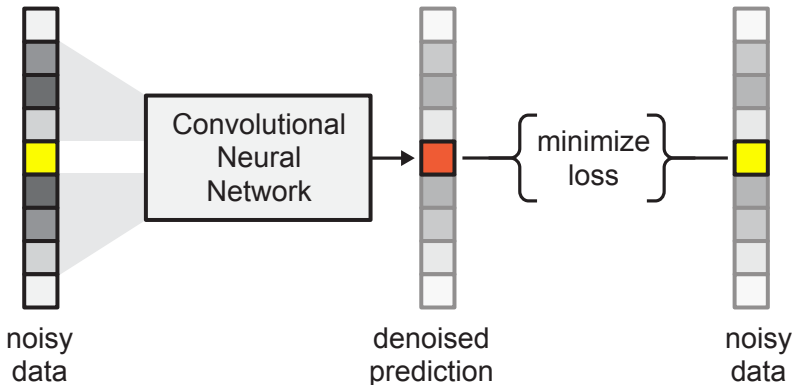
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Assumptions:

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Strategy 4: Blindspot denoising



Assumptions:

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Blindspot denoising

Two possible implementations:

Blindspot denoising

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1. Replace pixel by random noise in the input

Noise2Self: Blind denoising by self-supervision. Batson, Royer. 2019

Noise2Void - Learning denoising from single noisy images. Krull et al. 2019

Blindspot denoising

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Noise2Self: Blind denoising by self-supervision. Batson, Royer. 2019

Noise2Void - Learning denoising from single noisy images. Krull et al. 2019

2. Impose blindspot in the network architecture

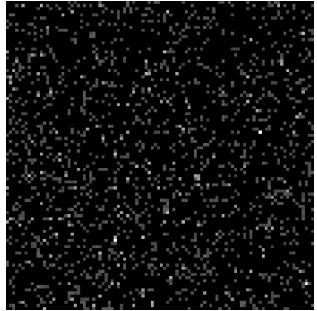
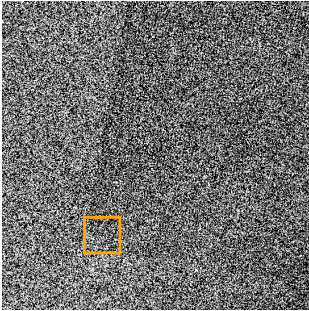
High-quality self-supervised deep image denoising. Laine et al. 2019

The Noise2Noise framework

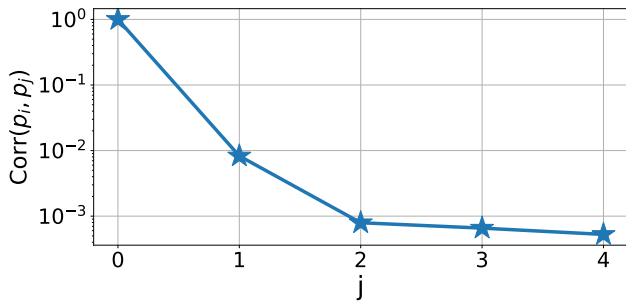
Unsupervised methods

Real data

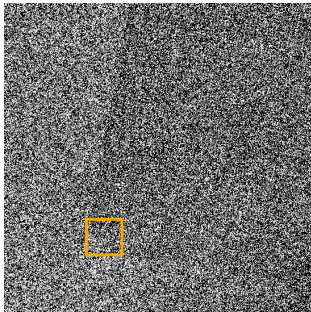
Electron microscopy



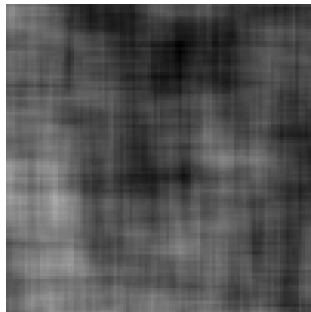
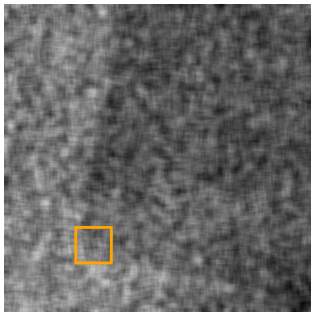
Spatial correlation of the noise



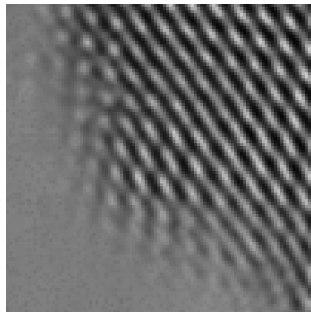
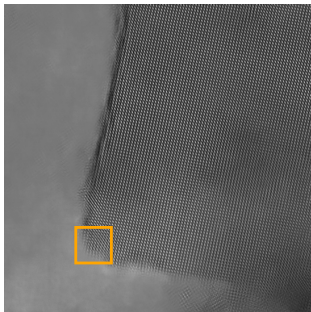
Noisy data



Gaussian smoothing, uPSNR: 20.4 dB

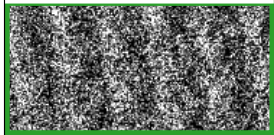
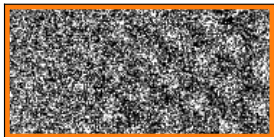
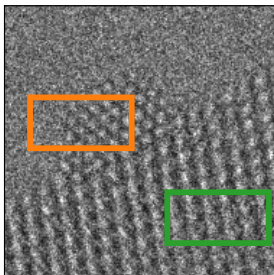


Neighbor2Neighbor, uPSNR: 26.9 dB



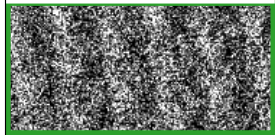
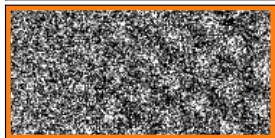
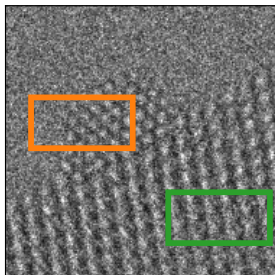
Blindspot denoising

Data

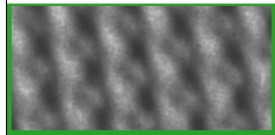
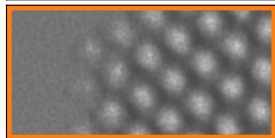
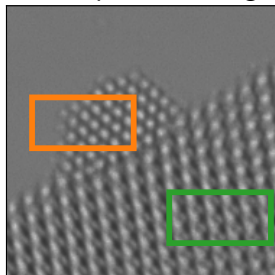


Blindspot denoising

Data



Blindspot denoising



Conclusions

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Key assumption: Noise independence

For more information

**Learn, Denoise and Discover: A Guide to Deep Denoising
with an Application to Electron Microscopy**

Sreyas Mohan, Kangning Liu, Peter Crozier,
Carlos Fernandez-Granda

Acknowledgements

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